

Understanding DNS Failures at Scale: Fine-grained Classification and Root Cause Analysis

Shaoxuan Yun
Beijing University of Posts and
Telecommunications
Beijing, China
shaouxuanyun@bupt.edu.cn

Yuchao Zhang
Beijing University of Posts and
Telecommunications
Beijing, China
yczhang@bupt.edu.cn

Yingqiang Wang
Computer Network Information
Center, Chinese Academy of Sciences
Beijing, China
yqwang@cnic.cn

Changhua Pei
Computer Network Information
Center, Chinese Academy of Sciences
Beijing, China
chpei@cnic.cn

Yi Li
Computer Network Information
Center, Chinese Academy of Sciences
Beijing, China
liyi@cstnet.cn

JingJing Li
Computer Network Information
Center, Chinese Academy of Sciences
Beijing, China
ljj@cnic.cn

Abstract

The Domain Name System (DNS) is fundamental to Internet infrastructure, and its stability is essential for ensuring reliable network access. While prior work has focused on coarse-grained failure statistics, in-depth analysis of specific failure types and their causes remains limited. In this study, we present a DNS failure analysis framework that combines coarse- and fine-grained perspectives over time. Using a real-world dataset of 2.5 billion DNS records, we systematically identify and classify major DNS failure patterns and quantify the distribution characteristics of each failure type, providing valuable insights into the underlying causes of DNS failures.

ACM Reference Format:

Shaoxuan Yun, Yuchao Zhang, Yingqiang Wang, Changhua Pei, Yi Li, and JingJing Li. 2025. Understanding DNS Failures at Scale: Fine-grained Classification and Root Cause Analysis. In *9th Asia-Pacific Workshop on Networking (AP-NET 2025)*, August 07–08, 2025, Shang Hai, China. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3735358.3737757>

1 Introduction

The Domain Name System (DNS) is a critical component of Internet infrastructure. Its failures can severely impact service accessibility and user experience [1]. While prior research has focused on high-level failure statistics, the lack of fine-grained analysis limits the understanding of specific failure behaviors and root causes [3]. To address this gap, we focus on analyzing the two most prevalent failure types, NXDOMAIN and NORESPONSE, observed in DNS queries of type A and AAAA. These failures collectively constitute the majority of DNS errors identified in real-world resolution logs collected from a large-scale public resolver. To conduct an in-depth analysis of DNS failures, we propose a hybrid analytical approach

combining coarse- and fine-grained perspectives to classify failure patterns and uncover their underlying causes.

2 DNS Failure Description

Table 1: The failure proportions of different response code.

RCODE Type	Statistical Result		Is Failure?
	Amount	Proportion	
NOERROR	1,967,656,917	72.4%	✗
NXDOMAIN	28,899,124	10.6%	✓
NORESPONSE	417,185,091	15.3%	✓
SERVFAIL	46,291,868	1.7%	✓
Total	2,720,125,700	100%	–

Table 2: The failure proportions of different request types.

Query type	A	AAAA	PTR	MX	other	Total
Failure Proportion	33.6%	57.8%	0.5%	0.1%	8.0%	100%

We analyze over 2.5 billion DNS resolution logs collected within a week from a major public DNS resolver in China, covering common query types like A and AAAA. To enable effective failure analysis, we design a hierarchical feature framework that captures both request-side attributes (e.g., query type) and response-side details (e.g., response codes).

To initiate our analysis, we first examine the DNS resolution failures to gain a high-level understanding of their distribution. Table 1 summarizes the distribution of DNS response codes (RCODEs) and indicates which of them are considered resolution failures. Notably, NXDOMAIN and NORESPONSE responses account for 10.6% and 15.3% of all queries, respectively, totaling a substantial 25.9%, while other failure types comprise only 1.7%. To further understand which types of DNS queries are most affected by these failures, we analyze the failure rates across different query types, and the results are presented in Table 2. The analysis reveals that DNS failures are primarily concentrated in queries for IPv4 addresses (A records, 33.6%) and IPv6 addresses (AAAA records, 57.8%). In

*Corresponding author.



This work is licensed under a Creative Commons Attribution International 4.0 License.

APNET 2025, Shang Hai, China

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1401-6/25/08

<https://doi.org/10.1145/3735358.3737757>

Table 3: Failure Overview

	NXDOMAIN ONLY		NORESPONSE ONLY	MIXED FAILURE		Total
	Illegal Domain	Non-existent Domain	Partially Serviceable	RFC Ambiguity	Service Offline	
Domain-level proportion	0.70%	89.26%	9.49%	0.014%	0.536%	100%
Request-level proportion	6.33%	36.97%	53.99%	0.21%	2.50%	100%

contrast, the failure rates for Pointer (PTR) and Mail Exchange (MX) queries are negligible, at only 0.5% and 0.1%, respectively. From this analysis, we derive a key observation: DNS failures are predominantly associated with A and AAAA queries, with NXDOMAIN and NORESPONSE being the most frequently observed response types. Accordingly, we focus on analyzing failures in A and AAAA queries that return NXDOMAIN and NORESPONSE. To facilitate subsequent experimental analysis, we extracted 619,856,543 such failure records, covering 15,625,665 unique domain names.

3 DNS Failure Analysis Method

To conduct an in-depth analysis of the underlying causes of DNS resolution failures mentioned in Section 2, we propose a DNS failure pattern analysis method that integrates both coarse-grained and fine-grained perspectives to systematically identify various failure patterns. Specifically, we first model the resolution behavior of each domain name based on the chronological order of requests and the corresponding RCODE. The model is formally defined as follows:

$$\text{Domain}_x = [(t_1, Q_1, R_1), (t_2, Q_2, R_2), \dots, (t_n, Q_n, R_n)] \quad (1)$$

Let Domain_x be the resolution log of the x -th domain, with t_i , Q_i , and R_i representing the timestamp, query type, and RCODE of the i -th request, respectively.

Then, based on the above resolution behavior model, we perform a coarse-grained classification of the domains into three categories: a) NXDOMAIN-only domains: Consistently return NXDOMAIN without NOERROR. b) NORESPONSE-only domains: Consistently return NORESPONSE with NOERROR. c) Mixed FAILURE domains: Exhibit varying failure types over time. Finally, building on this coarse-grained classification, we conduct a fine-grained analysis. First, we analyze intrinsic features of the domain name string, including its character composition (e.g., digits, hyphens, or unusual characters). Then, using failure timestamps and query types, we design automated scripts to characterize failure patterns from two dimensions: **a) Type Variability**: Analyzing the difference in RCODE between A and AAAA queries for the same domain. **b) Temporal Consistency**: Assessing whether failures are persistent (long-term) or sporadic (short-term). To validate our classification, we develop a multipoint verification system using the dig tool. This system periodically requeries failed domains through ten major global public DNS resolvers. By comparing current responses to historical logs, we evaluate the repeatability of failure patterns.

4 Results and Revealed Lessons

We applied the proposed analytical approach to the filtered dataset described in Section 2 and accordingly classified the failed domain names into five representative failure patterns. Table 3 presents the proportion of each failure pattern in terms of the number of domains and the corresponding request volume.

First, among domains that return only NXDOMAIN responses, a subset (accounting for 0.7% of the total failed domains) typically violates RFC naming conventions and lacks a valid structural format. These domains cannot be mapped to any real-world services, and the resulting 6.33% of failure requests essentially represent invalid entries within the DNS system. This reflects the fact that mainstream public DNS resolvers generally do not perform domain validity checks prior to processing queries, thereby allowing such invalid requests to propagate. Additionally, our dig-based probing reveals that the remaining NXDOMAIN-only domains do not exist at all, comprising 89.26% of total failed domains. We classify these two types of domains as *Illegal Domain* and *Non-existent Domain*, respectively.

Secondly, domains that return only NORESPONSE typically exhibit the following pattern: A record queries return NOERROR, whereas AAAA record queries result in NORESPONSE. Temporal analysis indicates that A and AAAA queries are usually issued concurrently, consistent with the behavior of the Happy Eyeballs algorithm. This algorithm is designed to facilitate the transition from IPv4 to IPv6 by simultaneously initiating both A and AAAA requests and preferring the address that is functional and responds more quickly. However, due to incomplete IPv6 deployment, many authoritative servers lack IPv6 addresses, resulting in frequent NORESPONSE outcomes for AAAA queries. This behavior is confirmed through our multipoint verification system. We refer to such domains as *Partially Serviceable*.

Finally, among the domains exhibiting mixed failure patterns, we observe that a subset (accounting for 0.014% of total failed domain) consistently returns NOERROR for A record queries, while AAAA record queries return NXDOMAIN. According to RFC 1035 [2], NXDOMAIN indicates that the domain and all its subdomains do not exist. Therefore, the coexistence of NXDOMAIN responses with successful A record resolutions violates the semantic rules defined by the RFC. We classify these as *RFC Ambiguity* domains. Additionally, we identify another subset of domains (accounting for 0.536% of the total failed domain) that initially returned NOERROR but gradually transitioned to persistent NXDOMAIN responses over time. This pattern reflects typical service offline behavior, which we define as the *Service Offline* domain.

5 Conclusion

In this paper, we propose a hybrid analysis approach that combines coarse-grained and fine-grained perspectives to investigate DNS resolution failures. By examining domain name structures, failure behaviors, and their temporal evolution, we conduct a fine-grained classification of failure patterns. Based on over 2.5 billion real-world DNS log entries, we identify five distinct failure modes and quantify their distributions. Our analysis also uncovers several practical

insights, revealing underlying causes—such as illegal domains and RFC ambiguities—that have been overlooked in prior research.

Acknowledgments

This work was supported by the National Key R&D Program of China under Grant 2024YFB2906900, the National Natural Science Foundation of China under Grant 62172054, the Beijing Nova Program under Grant 2023140, and the Research Project of Bytedance under Grant S2022054.

References

- [1] Guannan Liu, Lin Jin, Shuai Hao, Yubao Zhang, Daiping Liu, Angelos Stavrou, and Haining Wang. 2023. Dial" n" for nxdomain: The scale, origin, and security implications of dns queries to non-existent domains. In *Proceedings of the 2023 ACM on Internet Measurement Conference*. 198–212.
- [2] Paul V Mockapetris. 1987. Rfc1035: Domain names-implementation and specification.
- [3] Fenglu Zhang, Baojun Liu, Chaoyi Lu, Yunpeng Xing, Haixin Duan, Ying Liu, and Liyuan Chang. 2024. Investigating deployment issues of DNS root server instances from a China-wide view. *IEEE Transactions on Dependable and Secure Computing* (2024).